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United States Patent	12399870
Kind Code	B2
Date of Patent	August 26, 2025
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Data revision control in large-scale data analytic systems

Abstract

A method comprises storing first transaction data for a first transaction producing a first version of a first dataset, second transaction data for a second transaction transforming a first version of the first dataset to a second version of the first dataset, and third transaction data for a third transaction transforming the second version of the first dataset to a third version of the first dataset; storing dependency information indicating a first dependency of the third transaction on the second transaction and a second dependency of the second transaction on the first transaction; receiving a first instruction to revoke a first permission of a first user to access the second version of the first dataset; automatically revoking a certain permission of the first user to access the third version of the first dataset based on the dependency information.

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Appl. No.: 18/533003

Filed: December 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240104067 A1	Mar. 28, 2024

Related U.S. Application Data

continuation parent-doc US 17463345 20210831 US 11841835 child-doc US 18533003
continuation parent-doc US 16018777 20180626 US 11106638 20210831 child-doc US 17463345
continuation parent-doc US 15262207 20160912 US 10007674 20180626 child-doc US 16018777
us-provisional-application US 62349548 20160613

Publication Classification

Int. Cl.: **G06F16/18** (20190101); **G06F16/182** (20190101); **G06F16/21** (20190101); **G06F16/23** (20190101)

U.S. Cl.:

CPC **G06F16/1873** (20190101); **G06F16/182** (20190101); **G06F16/219** (20190101); **G06F16/2379** (20190101);

Field of Classification Search

CPC: G06F (16/1873); G06F (16/182); G06F (16/219); G06F (16/2379)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 120 as a Continuation of U.S. patent application Ser. No. 17/463,345 filed Aug. 31, 2021, which is a Continuation of U.S. patent application Ser. No. 16/018,777, filed Jun. 26, 2018, now U.S. Pat. No. 11,106,638, issued Aug. 31, 2021, which is a Continuation of U.S. patent application Ser. No. 15/262,207 filed Sep. 12, 2016, now U.S. Pat. No. 10,007,674, issued Jun. 26, 2018, which claims the benefit of U.S. Provisional Application 62/349,548, filed Jun. 13, 2016, the entire contents of all of which are hereby incorporated by reference as if fully set forth herein. The applicant hereby rescinds any disclaimer of scope in the parent applications or the prosecution history thereof and advises the USPTO that the claims in this application may be broader than any claim in the parent applications. (2) This application is related to U.S. patent application Ser. No. 14/533,433, filed Nov. 5, 2014, now U.S. Pat. No. 9,229,952, issued Jan. 5, 2016; and U.S. patent application Ser. No. 14/879,916, filed Oct. 9, 2015, now U.S. Pat. No. 9,483,506, issued Nov. 1, 2016, the entire contents of both of which are hereby incorporated by reference as if fully set forth herein.

TECHNICAL FIELD

(1) The disclosed implementations relate generally to large-scale data analytic systems. In particular, the disclosed implementations relate to data revision control in large-scale data analytic systems.

BACKGROUND

(2) Many large-scale data analytic systems are designed to efficiently run large-scale data processing jobs. For example, a traditional large-scale data analytic system is configured to execute large-scale data processing jobs on a cluster of commodity computing hardware. Such systems can typically execute job tasks in parallel at cluster nodes at or near where the data is stored, and aggregate and store intermediate and final results of task execution in a way that minimizes data movement between nodes, which would be expensive operationally given the large amount of data that is processed. Such systems also typically store data and job results in distributed file system locations specified by users but do not provide extensive revision control management of data and job results.

(3) Accordingly, the functionality of traditional large-scale data analytic systems is limited at least with respect to revision control of the data that is processed. Thus, there is a need for systems and methods that provide more or better revision control for data processed in large-scale data analytic systems. Such systems and methods may compliment or replace existing systems and methods for data revision control in large-scale data analytic systems.

SUMMARY

(4) The claims section at the end of this document provides a useful summary of some embodiments of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the various described implementations, reference should be made

to the Description of Implementations below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

(2) FIG. 1 is a data revision control model for large-scale data analytic systems.

(3) FIG. 2 illustrates a data revision control example according to the data revision control model.

(4) FIG. 3 is a block diagram of a data distribution network for data revision control in large-scale data analytic systems.

(5) FIG. 4 is a block diagram of an example large-scale data analytic system.

(6) FIG. 5 is a block diagram of a data revision control system.

(7) FIG. 6 is a very general block diagram of a computing device in which the example embodiment(s) of the present invention may be embodied.

(8) FIG. 7 is a block diagram of a basic software system for controlling the operation of the computing device.

(9) FIG. 8 is a flowchart of a process for determining column provenance.

(10) FIG. 9 illustrates a simple example of a build dependency graph, according to an embodiment of the present invention.

(11) FIG. 10 is a block diagram of a build catalog entry according to an embodiment of the present invention.

DESCRIPTION OF IMPLEMENTATIONS

(12) Reference will now be made in detail to implementations, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described implementations.

However, it will be apparent to one of ordinary skill in the art that the various described implementations may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the implementations.

(13) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first user interface could be termed a second user interface, and, similarly, a second user interface could be termed a first user interface, without departing from the scope of the various described implementations. The first user interface and the second user interface are both types of user interfaces, but they are not the same user interface.

(14) The terminology used in the description of the various described implementations herein is for the purpose of describing particular implementations only and is not intended to be limiting. As used in the description of the various described implementations and the appended claims, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(15) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting” or “in accordance with a determination that,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event]” or “in accordance with a determination that [a stated condition or event] is detected,” depending on the context.

Data Revision Control Model

(16) FIG. 1 is a block diagram of a data revision control model **100** for large-scale data analytic systems. The model **100** generally includes dataset versions **102**, transactions **104**, data files **106**, and driver programs **108**. Datasets are versioned in the context of transactions **104**. Specifically, each version **102** of a dataset corresponds to a different successfully committed transaction **104**. In the context of a transaction **104** that creates a new dataset version **102**, data may be added to a dataset if creating or revising the dataset and/or data may be removed from a dataset if revising the dataset. Data files **106** contain the data in datasets across dataset versions **102** including historical versions. Driver programs **108** are executed by large-scale data analytic systems (e.g., Apache Spark) in the context of transactions **104**. When executed, driver programs **108** apply parallel operations to one or more input dataset versions **102** and produce as a result one or more output dataset versions **102**.

(17) A simple example may be helpful to better understand the data revision control model **100**. FIG. 2 illustrates an example of data revision control according to data revision control model **100**.

(18) On Day One, an initial version of dataset A is created in the context of transaction TX1 resulting in data file F1. For example, data file F1 may contain web access log entries for the past six months. Also on Day One, an initial version of dataset B is created in the context of transaction TX2 resulting in data file F2. For example, data file F2 may contain rows corresponding to users of an online web service and associating user name identifiers with network addresses from which the users access the web service. Also on Day One, a driver program P1 is executed in the context of transaction TX3 that performs a join based on network address between dataset A, consisting of the initial version of dataset A, and dataset B, consisting of the initial version of dataset B. This execution results in an initial version of dataset C and data file F3 containing the results of the join operation executed in the context of transaction TX3.

(19) On Day Two, the previous day's (i.e., Day One's) web access log entries are added to dataset A in the context of transaction TX4 thereby producing data file F4. In this example, data file F4 contains only the previous day's (i.e., Day One's) web access log entries. Also on Day Two, the driver program P1 is executed again in the context of transaction TX5. In this example, the join performed in the context of transaction TX5 is between the web access log entries in data file F4 and the entries in data file F2. This execution results in a second version of dataset C and data file F5 containing the results of the join operation executed in the context of transaction TX5.

(20) Similarly, on Day Three, the previous day's (i.e., Day Two's) web access log entries are added to dataset A in the context of transaction TX6 and resulting in data file F6. In this example, data file F6 contains only the previous day's (i.e., Day Two's) web access log entries. Also on Day Two, the driver program P1 is executed again in the context of transaction TX7. In this example, the join performed in the context of transaction TX7 is between the web access log entries in data file F6 and the entries in data file F2. This execution results in a third version of dataset C and data file F7 containing the results of the join operation executed in the context of transaction TX7. As a result, there are three versions of dataset A corresponding to transactions TX1, TX4, and TX6 and data files F1, F4, and F6. There is one version of dataset B corresponding to transaction TX2 and data file F2. And there are three versions of dataset C corresponding to transactions TX3, TX5, and TX7 and data files F3, F5, and F7.

(21) While in this example and other examples presented herein there is a single data file **106** created for a dataset version **102** in the context of a transaction **104**, it is also possible for multiple data files **106** to be created for a dataset version **102**. Thus, a transaction **104** in which a dataset version **102** is created or revised may be associated with the more than one data file **106**.

Data Processing System

(22) In order to explain the operation of data revision control in large-scale data analytic systems, it is helpful to consider an exemplary distributed data processing system in which the data revision control is performed. In general, the implementations described here can be performed by a set of interconnected processors that are interconnected by one or more communication networks.

(23) FIG. 3 is a block diagram of an exemplary distributed data processing system **300**. It should be appreciated that the layout of the system **300** is merely exemplary and the system **300** may take on any other suitable layout or configuration. The system **300** is used to store data, perform computational tasks, and possibly to transmit data between datacenters. The system may include any number of data centers DCx, and thus the number of data centers shown in FIG. 3 is only exemplary. The system **300** may include dedicated optical links or other dedicated communication channels, as well as supporting hardware such as modems, bridges, routers, switches, wireless antennas and towers, and the like. In some implementations, the network **300** includes one or more wide area networks (WANs) as well as multiple local area networks (LANs). In some implementations, the system **300** utilizes a private network, e.g., the system and its interconnections are designed and operated exclusively for a particular company or customer. Alternatively, a public network may be used.

(24) Some of the datacenters may be located geographically close to each other, and others may be located far from the other datacenters. In some implementations, each datacenter includes multiple racks. For example, datacenter **302** includes multiple racks **308a**, . . . , **308n**. The racks **308** can include frames or cabinets into which components are mounted. Each rack can include one or more processors (CPUs) **310**. For example, the rack **308a** includes CPUs **310a**, . . . , **310n** (slaves **1-16**) and the nth rack **308n** includes multiple CPUs **310** (CPUs **17-31**). The processors **310** can include data processors, network attached storage devices, and other computer controlled devices. In some implementations, at least one of processors **310** operates as a master processor, and controls the scheduling and data distribution tasks performed throughout the network **300**. In some implementations, one or more processors **310** may take on one or more roles, such as a master and/or slave. A rack can include storage (e.g., one or more network attached disks) that is shared by the one or more processors **310**.

(25) In some implementations, the processors **310** within each rack **308** are interconnected to one another through a rack switch **306**. Furthermore, all racks **308** within each datacenter **302** are also interconnected via a datacenter switch **304**. As noted above, the present invention can be implemented using other arrangements of multiple interconnected processors.

(26) In another implementation, the processors shown in FIG. 3 are replaced by a single large-scale multiprocessor. In this implementation, data analytic operations are automatically assigned to processes running on the processors of the large-scale multiprocessor.

Example Large-Scale Data Analytic System

(27) In order to explain the operation of data revision control in large-scale analytic systems, it is also helpful to consider an exemplary large-scale data analytic system with which data revision control is performed. In general, the implementations described here can be performed by a cluster computing framework for large-scale data processing.

(28) FIG. 4 is a block diagram of an example large-scale data analytic system **400**. The system **400** provides data analysts with a cluster computing framework for writing parallel computations using a set of high-level operators with little or no concern about work distribution and fault tolerance. The system **400** is typically a distributed system having multiple processors, possibly including network attached storage nodes, that are interconnected by one or more communication networks. FIG. 4 provides a logical view of a system **400**, with which some implementations may be implemented on a system having the physical structure shown in FIG. 3. In one implementation, the system **400** operates within a single data center of the system **300** shown in FIG. 3, while in another implementation, the system **300** operates over two or more data centers of the system **300**.

(29) As shown in FIG. 4, a client **402** of a data analytic system **410** includes a driver program **404**. The driver program **404** is authored by a data analyst in a programming language (e.g., Java, Python, Scala, etc.) compatible with the data analytic system **410**. The driver program **404** implements a high-level control flow of an analytic application (e.g., text search, logistic regression, alternating least squares, interactive analytics, etc.) and launches various operations in parallel at a set of

worker machines **414**. The parallel operations operate on a set or sets of data distributed across the set of workers **414**.

(30) Generally, a set of distributed data operated on by the parallel operations is a collection of objects partitioned across the set of workers **414**. A set of distributed data may be constructed (instantiated) at the workers **414** from data in a data file **422** stored in a distributed file system cluster **420**. Alternatively, a set of distributed data can be constructed (instantiated) at the workers **414** by transforming an existing set of distributed data using a parallel transformation operation (map, filter, flatMap, groupByKey, join, etc.). A set of distributed data may also be persisted as a data file **422** to the distributed file system cluster **420** by a parallel save operation. Other parallel operations that may be performed at the workers **414** on a set of distributed data include, but are not limited to, reduce, collect, and for each. The reduce operation combines elements in a set of distributed data using associative function to produce a result at the driver program **404**. The collect operation sends all elements of a set of distributed data to the driver program **404**. The for each operation passes each element of a set of distributed data through a user provided function. Overall, executing the driver program **404** can involve constructing (instantiated) sets of distributed data at the set of workers **414** based on data read from data files **422**, constructing additional sets of distributed data at the set of workers **414** by applying transformation operations at the workers **414** to existing sets of distributed data, and persisting sets of distributed data at the workers **414** to data files **422** in the distributed file system cluster **420**.

(31) Cluster manager **412** provides a cluster operating system that lets the driver program **404** share the data analytic system cluster **410** in a fine-grained manner with other driver programs, possibly running at other clients. Cluster manager **412** also provides an application programming interface (API) invoke-able over a network by the driver program **404** via a network-based remote procedure call (RPC) protocol. In some implementations, the RPC protocol is, based on the Hyper Text Transfer Protocol (HTTP) or the Secure-Hyper Text Transfer Protocol (HTTPS). The cluster manager **412** API allows the driver program **404** to request task execution resources at the workers **414**. Generally, a task is a unit of work sent by the driver program **404** to an executor at a worker **414** for execution by the executor at the worker **414**. Generally, an executor is a process launched for the driver program **404** at a worker **414** that executes tasks **416** sent to it by the driver program **404**. The executor process runs tasks **416** and keeps data in memory or disk storage across tasks **416**. In some implementations, the driver program **404** is allocated dedicated executor processes at the workers **414** so that tasks **416** performed by the executor processes on behalf of the driver program **404** are process-isolated from tasks performed at the workers **414** on behalf of other driver programs.

(32) When an action (e.g., save, collect) is requested in the driver program **404**, the driver program **404** may spawn a parallel computation job. After spawning the job, the driver program **404** may then divide the job into smaller sets of tasks **416** called stages that depend on each other. The tasks **416** may then be scheduled according to their stages and sent to the executors allocated to the driver program **404** by the cluster manager **412** for execution at the workers **414**. Results of executing the tasks **416** at the workers **414** may be returned to the driver program **404** for aggregation and/or persisted to data files **422** in the distributed file system cluster **420**.

(33) The distributed data file system cluster **420** provides distributed data storage for data files **422** on a cluster of machines. The distributed data file system cluster **420** may present via an API a logical hierarchical file system to clients. With the cluster, data files **422** may be stored as data blocks distributed across machines of the cluster **420**. In some implementations, copies of data blocks are stored at different machines of the cluster **420** for fault tolerance and redundancy.

(34) The file system API for accessing, reading from, and writing to data files **422** may be invoke-able over a network from the client **402** including from the driver program **404** and from the workers **414** via a network-based remote procedure call (RPC) protocol. In some implementations, the RPC protocol is based on the HTTP or the HTTPS protocol. In some implementations, data

files **422** are identified via the API by Uniform Resource Identifiers (URIs). The URI for a data file **422** may comprise a scheme and a path to the data file **422** in the logical file system. In some implementations, the scheme is optional. Where a scheme is specified, it may vary depending on the type of cluster **420**. For example, if the cluster **420** is a Hadoop Distributed File System (HDFS) cluster, then the scheme of Mils for data files **422** may be “hdfs.” More generally, the API offered by the cluster **420** may supported accessing, reading from, and writing to data files **422** using any Hadoop API compatible URI.

Data Revision Control System

(35) FIG. 5 is a block diagram of a data revision control system **500**. The system **500** provides users of a large-scale data analytic system (e.g., system **400**) with a system to record data and to capture information about transformations that transform one piece of data into another piece of data.

(36) The system **500** includes a catalog service **512** that provides read and write access to a catalog **542** stored in a database **540**. Access to the catalog **542** by the catalog service **512** may be conducted in the context of transactions **104** supported by a database management system **530**.

(37) When access to a dataset version **102** is requested of the catalog service **512** by a user, the catalog service **512** may ask a permission service **514** if the user has permission to access the dataset version **102** according to dataset permissions **544** stored in the database **540** and accessible via the database management system **530**. If the user does not have access, then information in the catalog **542** such as transaction identifiers and file identifiers associated with the dataset **102** is not returned to the user.

(38) The user may interface with the catalog service **512** via the client **402**. The client **402** may be a command line-based or web-based. Via the client **402**, the user may request the catalog service **512** for a particular dataset version **102**, a particular transaction **104** of a dataset version **102**, or a particular file **106** of a dataset version **102**. If a particular dataset version **102**, then the catalog service **512**, assuming the user has permission to access the dataset version **102**, returns a set of paths to all data files **106** for all transactions **104** of the dataset version **102** recorded in the catalog **542**. If the request is for a particular transaction **104** of a dataset version **102**, then the catalog service **512**, again assuming the user has permission to access the dataset version **102**, returns a set of paths to all data files **106** for the transaction **104** recorded in the catalog **542**. If a particular data file **106** of a dataset version **102** is requested, then the catalog service **512**, once again assuming the user has permission to access the dataset version **102**, returns a path to the file **106** recorded in the catalog **542**.

(39) While in some implementations the user interfaces with the catalog service **512** and other services of the data revision control system **500** via a client specially configured to interface with services of the system **500**, the user interfaces with a service or services of the data revision control system **500** via a generic client (e.g., a standard web browser) in other implementations. Thus, there is no requirement that client **402** be specially configured to interface with network services of the data revision control system **500**.

(40) The client **402** may be coupled to a distributed file system **420** where the files **106** are actually stored. The client **402** may use file paths returned from the catalog service **512** to retrieve the bytes of the files **106** from the distributed file system **420**. The distributed file system **420** may be implemented the Hadoop Distributed File System (HFDS), Amazon S3 bucket, or the like.

(41) The catalog service **512** or the client **402** may request schema information **546** for a particular dataset version **102** or a particular file **106** of a dataset version **102** from the schema service **516**. The schema service **516** may verify that the requesting user has permission to the access the dataset version **102** first before providing the requested schema information to the catalog service **512** or the client **102**. The schema service **516** may retrieve the schema information from the database **540** via the database management system **530**.

(42) The catalog service **512** may manage encryptions keys for supporting file-level encryption of

files **106** stored in the distributed file system **420**. Specifically, the catalog **542** may store user-provided symmetric encryption keys in association with file identifiers of files **106** that are encrypted using the encryption keys. Provided the user has permission to access a requested dataset version **102**, the user-provided encryption keys may be returned to the client **402** along with the file paths in the catalog **542** to requested files **106** of the dataset **402**. The client **402** can decrypt the encrypted bytes retrieved from the distributed file system **420** using the user-provided encryption key for the file **106**. The user-provided encryption keys may be stored in the catalog **542** when the file **106** is initially created in the distributed file system **420**.

(43) The client **402** may be configured with an interface layer for processing user commands input via the command line or the web client and interacting with the catalog service **512**, the permission service **514**, the schema service **516**, and the distributed file system **420** to carry out those commands. For example, via the command line interface, the user may input a “change dataset” command to set the current dataset version **102** of the command line session (shell). Then the user may input a list command to obtain a list of transactions **104** or files **106** of the current dataset version **102**. The user may input a put command to add a specified file **106** to the dataset version **102**. Behind the scenes, the interface layer negotiates with the catalog service **512**, the permission service **514**, the schema service **516**, and the distributed file system **420** to carry out the commands.

(44) The interface layer may also exist on worker nodes **141** of the data analytic system cluster **410**. For example, the interface layer may also exist on Spark worker nodes such that when the worker nodes perform transformations **108** on dataset versions **102**, the interface layer negotiates with the services **512**, **514**, **516**, and/or **518** to facilitate the transformations **108**.

Catalog

(45) The data revision control system **500** may encompass maintaining an immutable history of data recording and transformation actions such as uploading a new dataset version **102** to the system **500** and transforming one dataset version **102** version to another dataset version **102**. The immutable history is referred to herein as the catalog **542**. The catalog **542** may be stored in a database **540**. Preferably, reads and writes from and to the catalog **542** are performed in the context of ACID-compliant transactions supported by a database management system **530**. For example, the catalog **542** may be stored in a relational database managed by a relational database management system that supports atomic, consistent, isolated, and durable (ACID) transactions. In one embodiment, the database management system **530** supporting ACID transactions is as described in related U.S. patent application Ser. No. 13/224,550, entitled “Multi-Row Transactions,” filed Sep. 2, 2011, the entire contents of which is hereby incorporated by referenced as if fully set forth herein.

Versioned Immutable Datasets

(46) The catalog **542** encompasses the notion of versioned immutable dataset versions **102**. More specifically, a dataset may encompass an ordered set of conceptual dataset items. The dataset items may be ordered according to their version identifiers recorded in the catalog **542**. Thus, a dataset item may correspond to a particular dataset version **102**. Or as another perspective, a dataset item may represent a snapshot of the dataset at a particular dataset version **102**.

(47) As a simple example, a version identifier of ‘1’ may be recorded in the catalog **542** for an initial dataset version **102**. If data is later added to the dataset, a version identifier of ‘2’ may be recorded in the catalog **542** for a second dataset version **102** that conceptually includes the data of the initial dataset version **102** and the added data. In this example, dataset version ‘2’ may represent the current dataset version **102** version and is ordered after dataset version ‘1’.

(48) As well as being versioned, a dataset version **102** may be immutable. That is, when a new dataset version **102** is created in the system **500**, pre-existing dataset versions **102** are not overwritten by the new dataset version **102**. In this way, pre-existing dataset versions **102** are preserved when a new dataset version **102** is added to a dataset. Note that supporting immutable dataset versions **102** is not exclusive of pruning or deleting dataset versions **102** corresponding to

old or unwanted dataset versions **102**. For example, old or unwanted dataset versions **102** may be deleted from the system **500** to conserve data storage space or in accordance with a data retention policy or regulatory compliance.

Dataset Transactions

(49) A dataset version **102** may correspond to a successfully committed transaction **104**. In these embodiments, a sequence of successfully committed transactions **104** may correspond to a sequence of dataset versions **102**.

(50) A transaction **104** against a dataset may add data to the dataset, edit existing data in the dataset, remove existing data from the dataset, or a combination of adding, editing, or removing data. A transaction **104** against a dataset may create a new dataset version **102** without deleting, removing, or modifying pre-existing dataset versions.

(51) A successfully committed transaction **104** may correspond to a set of one or more files **106** that contain the data of a dataset version **102** created by the successful transaction **104**. The set of files **106** may be stored in a file system **420**. In a preferred embodiment, the file system **420** is the Hadoop Distributed File System (HDFS) or other distributed file system **420**. However, a distributed file system **420** is not required and a standalone file system may be used.

(52) In the catalog **542**, a dataset version **102** may be identified by the name or identifier of the dataset version **102**. In a preferred embodiment, the dataset version **102** corresponds to an identifier assigned to the transaction **104** that created the dataset version **102**. The dataset version **102** may be associated in the catalog with the set of files **106** that contain the data of the dataset version **102**. In a preferred embodiment, the catalog **542** treats the set of files **106** as opaque. That is, the catalog **542** itself may store paths or other identifiers of the set of files **106** but may not otherwise open, read, or write to the files **106**.

(53) In sum, the catalog **542** may store information about dataset versions **102**. The information may include information identifying different dataset versions **102**. In association with information identifying a particular dataset version **102**, there may be information identifying one or more files **106** that contain the data of the particular dataset version **102**.

Dataset Branching

(54) The catalog **542** may store information representing a non-linear history of a dataset. Specifically, the history of a dataset may have different dataset branches. Branching may be used to allow one set of changes to a dataset to be made independent and concurrently of another set of changes to the dataset. The catalog **542** may store branch names in association with identifies of dataset versions **102** for identifying dataset versions **102** that belong to a particular dataset branch.

Dataset Provenance

(55) The catalog **542** may provide dataset provenance at the transaction level of granularity. As an example, suppose a driver program **108** is executed in the data analytic system **400** multiple times that reads data from a version of dataset A, reads data from a version of dataset B, transforms the data from the version of dataset A and the data from the version of dataset B in some way to produce a version of dataset C. As mentioned, this transformation may be performed multiple times. Each transformation may be performed in the context of a transaction **104**. For example, the transformation may be performed daily after datasets A and B are updated daily in the context of transactions. The result being multiple versions of dataset A, multiple versions of dataset B, and multiple versions of dataset C as a result of multiple executions of the driver program **108**. The catalog **542** may contain sufficient information to trace the provenance of a particular version of dataset C to the versions of datasets A and B from which the particular version of dataset C is derived. In addition, the catalog may contain sufficient information to trace the provenance of those versions of datasets A and B to the earlier versions of datasets A and B from which those versions of datasets A and B were derived.

(56) The provenance tracking ability is the result of recording in the catalog **542** for a transaction **104** that creates a new dataset version **102**, the transaction **104** or transactions **104** that the given

transaction **104** depends on (e.g., is derived from). The information recorded in the catalog **542** may include an identifier of each dependent transaction **104** and a branch name of the dataset version **102** that the dependent transaction **104** was committed against.

(57) According to some embodiments, provenance tracking extends beyond transaction level granularity to column level granularity. For example, suppose a dataset version **A1** is structured as a table of two columns and a dataset version **B1** is structured as a table of five columns. Further assume, column three of dataset version **B1** is computed from column one of dataset version **A1**. In this case, the catalog may store information reflecting the dependency of column three of dataset version **B1** on column one of dataset version **A1**.

Permission Transitivity

(58) The catalog **542** may also support the notion of permission transitivity. For example, suppose the catalog **542** records information for two transactions **104 2** referred to in this example as “Transaction **1**” and Transaction **2**.” Further suppose a third transaction **104**, referred to in this example as “Transaction **3**,” is performed against a dataset version **102**. Transaction **3** may use data created by Transaction **1** and data created by Transaction **2** to create a new dataset version **102**. After Transaction **3** is executed, it may be decided according to organizational policy that a particular user should not be allowed to access the data created by Transaction **2**. In this case, as a result of the provenance tracking ability, and in particular because the catalog **542** records the dependency of Transaction **3** on Transaction **2**, if permission to access the data of Transaction **2** is revoked from the particular user, permission to access the data of Transaction **3** may be transitively revoked from the particular user.

(59) The transitive effect of permission revocation (or permission grant) can apply to an arbitrary number of levels in the provenance tracking. For example, returning to the above example, permission may be transitively revoked for any transaction that depends directly or indirectly on the Transaction **3**.

(60) According to some embodiments, where provenance tracking in the catalog **542** has column level granularity. Then permission transitivity may apply at the more fine-grained column-level. In this case, permission may be revoked (or granted) on a particular column of a dataset version **102** and based on the column-level provenance tracking in the catalog **542**, permission may be transitively revoked on all direct or indirect descendent columns of that column.

Build Service

(61) The system **500** may encompass a build service **518** that manages driver programs **108** which are executed in the analytic system **400** to transform data. The build service **518** may leverage a directed acyclic graph data structure to ensure that driver programs **108** are executed in proper dependency order. The graph may encompass a node representing an output dataset to be computed based on one or more input datasets each represented by a node in the graph with a directed edge between node(s) representing the input dataset(s) and the node representing the output dataset. The build service **518** traverses the DAG in dataset dependency order so that the most upstream dependent datasets are computed first. The build service traverses the DAG from the most upstream dependent datasets toward the node representing the output dataset rebuilding datasets as necessary so that they are up-to-date. Finally, the target output dataset is built once all of the dependent datasets are up-to-date.

(62) The build service **518** may maintain build dependency data **548** representing one or more directed acyclic graphs (also referred to herein as a “build dependency graph”). There may be multiple such graphs if, for example, none of the datasets represented by a graph has a build dependency on a dataset represented by another graph. Each graph comprises nodes and one or more directed edges connecting the nodes. A leaf node of a graph corresponds to a dataset that does not have any build dependencies on another dataset. A non-leaf node of a graph corresponds to a dataset that has a build dependency on at least one other dataset. A root node of a graph is a non-leaf node but where there are no build dependencies on the dataset represented by the root node. A

graph may have only one root node or may have multiple root nodes. A directed edge connecting two nodes in a graph represents a build dependency between two datasets. A graph may be represented in a computer memory as an N-ary tree data structure or other suitable data structure. (63) To illustrate a build dependency graph by a simple example, consider graph **900** of FIG. **9**. Each circle of graph **900** represents a node of the build dependency graph and each arrow connecting two circles of graph **900** represents a directed edge of the build dependency graph. The letter in each circle of graph **900** represents the name of the dataset represented by the corresponding node. As shown, datasets F and A are represented by root nodes of the build dependency graph, datasets C, D, and E are represented by leaf nodes of the build dependency graph, and dataset B is represented by a non-leaf node of the build dependency graph. Also shown, dataset F has a build dependency on dataset C, dataset B has build dependencies on datasets C and D, and dataset A has build dependencies on datasets B, C, D, and E. Dataset A's build dependency on dataset C is transitive by way of dataset B. Datasets F and B may be considered the “parent” datasets of dataset C (and dataset C the “child” of datasets F and B), datasets B and A the parent datasets of dataset D (and dataset D the child of datasets B and A), and dataset A the parent dataset of datasets B, D, and E (and datasets B, D, and E the children of dataset A). However, dataset A is not considered a parent of dataset C and dataset C is not considered a child of dataset A.

(64) To determine whether a given dataset is out-of-date with respect to its build dependencies, the build service **518** traverses the build dependency graph starting at the node corresponding to the given dataset and visits at least every non-leaf node in the graph sub-tree that is rooted at the node corresponding to the given dataset. During the traversal, nodes are visited in post-order according to a depth-first traversal algorithm. For example, referring briefly to FIG. **9**, if the given dataset is A, then a post-order depth-first recursive traversal of graph **900** would visit the node for dataset C and the node for dataset D before visiting the node for dataset B and would visit the node for dataset D and the node for dataset E before visiting the node for dataset A.

(65) For each non-leaf node visited during the traversal, a determination is made whether the current version of the dataset corresponding to the visited non-leaf node is out-of-date with respect to any of its child datasets. To determine whether the current version of a dataset is out-of-date with respect to a child dataset, the build service **518** may consult the catalog **542** by way of the catalog service **512**. If the current version of the dataset is out-of-date with respect to any of its child datasets, then the build service **518** executes the current version of the driver program for the dataset to generate a new version of the dataset. After the new version of the dataset has been generated, the build service **518** adds a new entry to the catalog **542** reflecting the new version of the dataset. In an embodiment, datasets are recursively rebuilt if dependencies of the dataset to be rebuilt are also out-of-date.

Build Service Branches

(66) Build service branches allow the same driver program **108** to be executed on multiple branches. For example, a driver program **108** on the master branch can be executed to produce a dataset version **102** on the master branch or on another branch (e.g., the develop branch). Build server branches also allow a driver program **108** on a branch to be executed to produce dataset versions **102** on that branch. For example, a driver program **108** on the develop branch can be executed to produce a dataset version **102** that is available only on the develop branch. Build server branches provide isolation of re-computation of graph data across different users and across different execution schedules of a data pipeline. To support branching, the catalog **542** may store information represents a graph of dependencies as opposed to a linear dependency sequence.

Build Fallbacks

(67) The build service may support build fallbacks between branches. For example, suppose the system stores two “raw” datasets A and B that are both updated daily (e.g., with daily web log data for two web services). Each update creates a new version **102** of the dataset and corresponds to a different transaction **104**. The datasets are deemed raw in the sense that a driver program **108** is not

executed by the analytic system **400** to produce versions **102** of the datasets versions of the datasets.

(68) Further suppose there is a driver program **108** that computes a join between datasets A and B. The join may be performed in a data transformation system such a SQL database system or Apache Spark, for example. More generally, the techniques described herein are agnostic to the particular analytic system that is used. The data to be transformed and the driver program **108** to transform the data can be provided to the analytic system **400** based on information stored in the catalog **542** including where to store the output data.

(69) If the driver program **P1** is executed on the master branch, it may produce an output dataset version **102** also on the master branch. Here, the dataset version **102** may be considered a “derived” dataset version **102** because it is derived from one or more other dataset versions **102**. Further suppose there is another driver program **P2** on the master branch that filters the data in a dataset version **102** and produces a filtered dataset version **102**, also on the master branch. Now suppose a new branch called “develop” is created that includes a modification of driver program **P2**. The modification is stored in the system **500** as a new version of the driver program **P2** on the develop branch. For example, the new version of the driver program **P2** may apply the filter operation of the prior version of the driver program **P2** in a different way.

(70) When requested to build a target branch that is branched from another parent branch, the build service **518** may fall back to the parent branch when a necessary input dataset version **102** is not present on the target branch. For example, if requested to bring the develop branch up to date in the example above, the build service **518** may fallback to providing a dataset version **102** on the master branch as input to a version of a driver program **108** on the develop branch to produce a filtered dataset version **102** on the develop branch.

(71) The catalog **542** may store parent-child and fallback relationships between branches. For example, the develop branch in the example above is a child of the master branch. Another branch, for example an “experimental” branch, may be a further child of the develop branch and so on. As in the above example, when the build service **518** is requested to build a branch that is a child of another branch, the build service **518** may identify a parent dataset version **102** or a parent version of a driver program **108** from a parent branch if no parent version of a dataset or no parent version of the driver program **108** exists on the target child branch. This identification may be recursive if the identification cannot be made on the parent branch and the parent branch is a child branch of another branch and so on.

(72) Thus, the build service **518** supports GIT-style branching for both data and code.

Builds

(73) In a command to the build service **518** to build a branch, the build service may accept as a parameter to the command, a target driver program **108**. For example, in an example pipeline on the master branch involving datasets A, B, C, and D and driver programs **P1** and **P2** where driver program **P1** builds the current versions of dataset C from the current versions of datasets A and B and driver program **P2** builds the current version of dataset D from the current version of dataset C, the driver programs **P1** or **P2** may be specified as the target. For example, if driver program **P2** is specified as the target, then the build service may read and process data in the catalog **542** to determine whether a transaction **104** updating any of dataset A, dataset B, and driver program **P1** has committed since the transaction **104** produced the current version of dataset C was committed. If not, then dataset C is up-to-date, otherwise dataset C is out-of-date. If dataset C is out-of-date, then the build service rebuilds dataset C to create a new current version of dataset C based on the current versions of datasets A and B and the current version of driver program **P1**. If dataset C is already up-to-date or after dataset C is brought up-to-date, the build service generates a new version of dataset D based on the now current version of dataset C and driver program **P2**.

Push Build

(74) According to some embodiments, the build service **518** supports a push build. In a push build,

rebuilt of all datasets that depend on an upstream dataset or an upstream driver program **108** that has been updated are automatically determined based on information in the catalog **542** and rebuilt. In this case, the build service **518** may accept a target dataset or a target driver program **108** as an input parameter to a push build command. The build service **518** then determines all downstream datasets that need to be rebuilt, if any.

(75) As an example, if the build service **518** receives a push build command with dataset A as the target, then the build service would determine all downstream datasets that are not up-to-date with respect to dataset A and rebuild them. For example, if dataset C is out-of-date with respect to dataset A, then dataset C is rebuilt based on the current versions of datasets A and B and the current version of driver program **P1**. If dataset C is rebuilt because it is out-of-date, then dataset D will be rebuilt based on the up-to-date version of dataset C and the current version of driver program **P2** and so on until all downstream datasets of the target dataset are rebuilt. The build service **518** may perform similar rebuilding if the target of the push build command is a driver program **108**.

Triggers

(76) The build service **518** may also support triggers. In this case, a push build may be considered a special case of a trigger. A trigger, generally, is an action (e.g., a rebuild) that is performed by the build service **518** that is triggered by the creation of a new version **102** of a dataset or a new version of a driver program **108** in the system **500**.

Build Catalog

(77) According to some embodiments of the present invention, builds data **548** may encompass a build catalog for tracking builds orchestrated by the build service **514**. In an embodiment, as exemplified in FIG. **10** an entry in the build catalog corresponding to a non-leaf node in a build dependency graph may comprise a dataset name **1002**, a dataset version **1004**, and build dependency information **1006**. An entry may also comprise a build branch name if the target of a build is a particular branch. Build service **518** adds a new build catalog entry to the build catalog each time a new version of a dataset is built and committed in the context of a transaction **104** facilitated by the build service **518**. Thus, the build catalog may store a build catalog entry for each version of a dataset, including the current version of the dataset and any historical (prior) versions of the dataset.

(78) The dataset name **1002** may be a unique identifier of a dataset. The dataset name **1002** may be used to identify the dataset across all versions of the dataset. In other words, the dataset name **1002** may be the same in all build catalog entries for all versions of the dataset.

(79) The dataset version **1004** is a unique identifier of a version of the dataset. Typically, the dataset version **1004** is an ordinal or other information that can be used to determine whether the version of the dataset represented by the dataset version **1004** happened before or happened after other versions of the dataset represented by other build catalog entries in the build catalog with the same dataset name **1002**.

(80) In an embodiment, the dataset version **1004** is an identifier (e.g., a transaction commit identifier) assigned by the DBMS **530** to a commit of a transaction that stored the version **1004** of the dataset.

(81) The build dependencies **1006** may comprises a list of one or more dataset build dependencies **1008** and a driver program build dependency **1010**. The list of dataset build dependencies **1008** correspond to any child datasets input to the version of the derivation program used to build the version **1004** of the derived dataset. If no such datasets were input, then the list of dataset build dependencies **1008** may be an empty list.

(82) In an embodiment, each dataset build dependency (e.g., **1008A**) specifies the name and the version of a dataset that the version **1004** of the derived dataset was built (generated) from. For example, the name and the version of a dataset build dependency (e.g., **1008B**) may correspond to a dataset name **1002** and a dataset version **1004** of a build catalog entry for a version of a dataset that the version **1004** of the derived dataset was generated (built) from.

(83) In an embodiment, the derivation program build dependency **1010** specifies the name and the version of a driver program **108** that the build service **518** executed to generate (build) the version **1004** of the derived dataset. For example, the name and the version of the derivation program dependency **1010** may correspond to a derivation program entry for the version of the derivation program that was executed by the build service **518** to generate (build) the version **1004** of the derived dataset.

(84) In an embodiment, the build service **518** identifies the current version of a derived dataset by querying build catalog entries for the build catalog entry comprising the latest (most recent) dataset version **1004** and having a dataset name **1002** matching a name for the derived dataset specified as a query parameter.

(85) In an embodiment, the build service **518** determines whether the current version of a given dataset is out-of-date based on the build catalog entry for the current version of the given dataset. The current version of the given dataset may be considered out-of-date for any one of a number of reasons including because: 1) there is a version of the driver program **108** that is newer than the version used to build the current version of the given dataset, 2) there is a version of a child dataset that is newer the version of the child dataset from which the current version of the given dataset was built, or 3) a dependency of the given dataset on another dataset was added or removed.

Schema Metadata Service

(86) The system **500** encompass a schema metadata service **516** that stores schema information about files **106** that correspond to transactions **104** reflected in the catalog **542**. The schema information may encompass metadata related to a dataset, dataset version, a transaction, or a branch. An identifier of a given file **106** identified in the catalog **542** may be passed to the schema metadata service **516** and the schema metadata service **516** may return schema information for the file **106**. The schema information may encompass data schema related information such as whether the data in the file **106** is structured as a table, the names of the columns of the table, the data types of the columns, user descriptions of the columns, etc.

(87) The schema information accessible via the schema metadata service **516** may versioned separately from the data itself in the catalog **542**. This allows the schemas to be updated separately from datasets **102** and those updates to be tracked separately. For example, suppose a comma separated file is uploaded to the system **500** as particular dataset version. The catalog **542** may store in association with the particular dataset version identifiers of one or more files **106** in which the CSV data is stored. The catalog **542** may also store in association with each of those one or more file identifiers, schema information describing the format and type of data stored in the corresponding file **106**. The schema information for a file **106** may be retrievable via the schema metadata service **516** given an identifier of the file **106** as input. Note that this versioning scheme in the catalog **542** allows new schema information for a file **106** to be associated with the file **106** and accessible via the schema metadata service **516**. For example, suppose after storing initial schema information for a file **106** in which the CSV data is stored, updated the schema information is stored that reflects a new or better understanding of the CSV data stored in the file **106**. The updated schema information may be retrieved from the schema metadata service for the file **106** without having create a new version of the CSV data or the file **106** in which the CSV data is stored.

(88) When a transformation **108** is executed, the build service **516** may encapsulate the complexities of the separate versioning of datasets and schema information. For example, suppose driver program **P1** accepts dataset A and dataset B as input is the target of a build command issued to the build service **518**. In response to this build command, the build service **518** may determine from the catalog **542** the file **106** or files **106** in which the data of the current versions of datasets A and B is stored. The build service **518** may then access the schema metadata service **516** to obtain the current versions of the schema information for the file **106** or files **106**. The build service **518** may then provide all of identifiers or paths to the file **106** or files **106** and the obtained schema

information to the analytic system **400** (e.g., Apache Spark) to execute the driver program **P1**. The underlying data transformation engine interprets the schema information and applies it to the data in the file **106** or files **106** when executing the transformation A.

Column Provenance

(89) According to some embodiments, the system **500** supports column provenance. FIG. **8** is a flowchart of a process **800** for determining column provenance according to some embodiments of the present invention.

(90) At step **802**, a driver program is Obtained. Specifically, the instructions of the driver program are parsed.

(91) At step **804**, based on the parsing of the driver program, one or more input datasets and the output dataset are identified. The input dataset(s) are the datasets that the driver program will consume data from and apply transformations to in producing the output dataset.

(92) At step **806**, the schema(s) for the input dataset(s) and the output dataset are obtained. For example, the schema for an input dataset or the output dataset may be retrieved from the schema service based on an identifier of the dataset.

(93) At step **808**, based on the parsing of the driver program, one or more transaction operations are identified in the driver program.

(94) At step **810**, for each transformation operation identifier, it is determined, based on the obtained schemas, whether the transformation operation adds a column, removes a column, or updates a column in the output dataset.

(95) At step **812**, for each “target” column added to, removed from, or updated in the output dataset, column provenance information for the target column is determined and recorded in a database. The column provenance information for the target column may include the name or identifier of the target column that is added, removed, or updated. In addition, if the target column is added or updated and the values in the added or updated column are based on applying a transformation to values in one or more “dependent” columns (e.g., a join), then the column provenance information for the target column includes a link between the target column and the one or more dependent columns.

(96) At step **814**, a column provenance graph is generated and displayed in a user interface based on the recorded column provenance information. The graph may include nodes and edges. The nodes may represent columns that have been added or updated. The edge may represent dependencies between target columns and dependent columns. The presented graph is useful to user for debugging how a transformation changed a dataset including erroneous changes to a dataset. For example, the presented graph may be useful for easily tracking down a column of dataset that incorrectly has all NULL values. This is because the graph can show by nodes and edges the column provenance of the incorrect column. The graph also allows multiple users working on different driver programs to see how their work affects one another and possibly identify dependencies between different users' driver programs.

Column Provenance Graph

(97) According to some embodiments, a column provenance graph is generated and displayed in graphical user interface to a user of the system **500**. As mentioned, the graph comprises nodes and edges connected the nodes. The edges may be directed and may represent a transformation of the values in a “source” column to produce values for a “target” column. The edge may be directed from the source column node to the target column node to represent that the values in the source column were transformed to produce the values in the target column. Alternatively, the edge may be directed from the target column node to the source column node to represent that the values in the target column were derived from a transformation applied to values in the source column.

(98) According to some embodiments, a directed edge corresponds to one or more transformation operations in a driver program. According to some embodiments, a user can interact with a directed edge (e.g., click on or touch the directed edge) via the graphical user interface to reveal (e.g., in a

pop-up box or other graphical user interface) the transformation operations that are represented by the directed edge.

(99) According to some embodiments, each node of a column provenance graph corresponds to either a starting column from an input dataset to a driver program, an intermediate column that is not a final output column but that is produced by applying a transformation to a starting column or another intermediate column, or a target column of the target output dataset that is produced by applying a transformation to a starting column or an intermediate column.

Dataset Provenance Graph

(100) According to some embodiments, a dataset provenance graph is generated and displayed in graphical user interface to a user of the system **500**, in addition to or instead of generation and display of a column provenance graph. The dataset provenance graph comprises nodes and directed edges. Each of the nodes represents a dataset and edges represent derivation dependencies between dataset. For example, a directed edge from dataset A to dataset B represents that data in dataset B is derived from data in dataset A.

(101) According to some embodiments, the nodes representing datasets are color coded in the dataset graph to indicate that they are out-of-date with respect to a direct parent dataset in the graph. For example, if data is added to dataset A after the most recent time dataset B was derived from dataset A, then the node representing dataset B may be color-coded (e.g., in red) to indicate that dataset B is out-of-date with respect dataset A.

(102) According to some embodiments, a directed edge between two nodes representing two datasets in the dataset provenance graph represents a dependency between columns of the two datasets. For example, suppose data in column C1 in dataset A is derived from data in column C2 in dataset B and data in column C3 in dataset C. In this case, a first directed edge may be drawn in the dataset provenance graph from the node representing dataset A to the node representing dataset B and a second directed edge may be drawn in the graph from the node representing dataset A to the node representing dataset C based on determining the derivation dependencies of column C1 of dataset A on columns C2 and C3 of datasets B and C, respectively.

(103) According to some embodiments, the dataset provenance graph allows users to add annotations to nodes and edges of the graph. The annotations can be in the form of textual metadata that is associated with a selected node or edge. The annotations may be visible by other users of the system **500**, for example, by clicking on or hovering over or otherwise selecting a node or edge in the graph that is associate with an annotation. The annotations provide a way for users to document the graph for other users with useful information.

(104) According to some embodiments, the user can select one or more nodes in the dataset provenance graph corresponding to one or more datasets that that user wants to refresh. In this context, a refresh of a “target” dataset refers to executing a driver program to rebuild the target dataset based on the latest data available in the “source” datasets on which the target dataset depends according to the graph.

(105) According to some embodiments, if a dataset the user selects as a target is not out-of-date with respect to an ancestor dataset, then the user is prompted to confirm that the target dataset should be rebuilt. The dataset is rebuilt if the user confirms the refresh.

(106) According to some embodiments, if a dataset that is an ancestor to a target dataset the user selects to refresh is out-of-date but the ancestor dataset is not included in the dataset(s) the user selects to refresh, then the user is prompted to add the ancestor dataset to the set of datasets to refresh.

(107) According to some embodiments, the user may enter a parameter that specifies the maximum number of driver programs to execute concurrently on the data analytic cluster **400** so as to avoid overloading the cluster with computation jobs.

(108) According to some embodiments, before any driver programs are scheduled for execution on the analytic cluster **400**, the driver programs for the target refresh datasets are compiled for

compilation errors (e.g., syntax errors). If any of the driver programs fail compilation, then none of the driver programs are executed so to provide data integrity and to catch potential refresh errors before a time consuming rebuild is initiated.

Basic Computing Device

(109) Referring now to FIG. 6, it is a block diagram that illustrates a basic computing device **600** in which the example embodiment(s) of the present invention may be embodied. Computing device **600** and its components, including their connections, relationships, and functions, is meant to be exemplary only, and not meant to limit implementations of the example embodiment(s). Other computing devices suitable for implementing the example embodiment(s) may have different components, including components with different connections, relationships, and functions.

(110) Computing device **600** may include a bus **602** or other communication mechanism for addressing main memory **606** and for transferring data between and among the various components of device **600**.

(111) Computing device **600** may also include one or more hardware processors **604** coupled with bus **602** for processing information. A hardware processor **604** may be a general purpose microprocessor, a system on a chip (SoC), or other processor.

(112) Main memory **606**, such as a random access memory (RAM) or other dynamic storage device, also may be coupled to bus **602** for storing information and software instructions to be executed by processor(s) **604**. Main memory **606** also may be used for storing temporary variables or other intermediate information during execution of software instructions to be executed by processor(s) **604**.

(113) Software instructions, when stored in storage media accessible to processor(s) **604**, render computing device **600** into a special-purpose computing device that is customized to perform the operations specified in the software instructions. The terms “software”, “software instructions”, “computer program”, “computer-executable instructions”, and “processor-executable instructions” are to be broadly construed to cover any machine-readable information, whether or not human-readable, for instructing a computing device to perform specific operations, and including, but not limited to, application software, desktop applications, scripts, binaries, operating systems, device drivers, boot loaders, shells, utilities, system software, JAVASCRIPT, web pages, web applications, plugins, embedded software, microcode, compilers, debuggers, interpreters, virtual machines, linkers, and text editors.

(114) Computing device **600** also may include read only memory (ROM) **608** or other static storage device coupled to bus **602** for storing static information and software instructions for processor(s) **604**.

(115) One or more mass storage devices **610** may be coupled to bus **602** for persistently storing information and software instructions on fixed or removable media, such as magnetic, optical, solid-state, magnetic-optical, flash memory, or any other available mass storage technology. The mass storage may be shared on a network, or it may be dedicated mass storage. Typically, at least one of the mass storage devices **610** (e.g., the main hard disk for the device) stores a body of program and data for directing operation of the computing device, including an operating system, user application programs, driver and other support files, as well as other data files of all sorts.

(116) Computing device **600** may be coupled via bus **602** to display **612**, such as a liquid crystal display (LCD) or other electronic visual display, for displaying information to a computer user. In some configurations, a touch sensitive surface incorporating touch detection technology (e.g., resistive, capacitive, etc.) may be overlaid on display **612** to form a touch sensitive display for communicating touch gesture (e.g., finger or stylus) input to processor(s) **604**.

(117) An input device **614**, including alphanumeric and other keys, may be coupled to bus **602** for communicating information and command selections to processor **604**. In addition to or instead of alphanumeric and other keys, input device **614** may include one or more physical buttons or switches such as, for example, a power (on/off) button, a “home” button, volume control buttons,

or the like.

(118) Another type of user input device may be a cursor control **616**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **604** and for controlling cursor movement on display **612**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane.

(119) While in some configurations, such as the configuration depicted in FIG. **6**, one or more of display **612**, input device **614**, and cursor control **616** are external components (i.e., peripheral devices) of computing device **600**, some or all of display **612**, input device **614**, and cursor control **616** are integrated as part of the form factor of computing device **600** in other configurations.

(120) Functions of the disclosed systems, methods, and modules may be performed by computing device **600** in response to processor(s) **604** executing one or more programs of software instructions contained in main memory **606**. Such software instructions may be read into main memory **606** from another storage medium, such as storage device(s) **610**. Execution of the software instructions contained in main memory **606** cause processor(s) **604** to perform the functions of the example embodiment(s).

(121) While functions and operations of the example embodiment(s) may be implemented entirely with software instructions, hard-wired or programmable circuitry of computing device **600** (e.g., an ASIC, a FPGA, or the like) may be used in other embodiments in place of or in combination with software instructions to perform the functions, according to the requirements of the particular implementation at hand.

(122) The term “storage media” as used herein refers to any non-transitory media that store data and/or software instructions that cause a computing device to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, non-volatile random access memory (NVRAM), flash memory, optical disks, magnetic disks, or solid-state drives, such as storage device **610**. Volatile media includes dynamic memory, such as main memory **606**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid-state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, flash memory, any other memory chip or cartridge.

(123) Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **602**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(124) Various forms of media may be involved in carrying one or more sequences of one or more software instructions to processor(s) **604** for execution. For example, the software instructions may initially be carried on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the software instructions into its dynamic memory and send the software instructions over a telephone line using a modem. A modem local to computing device **600** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **602**. Bus **602** carries the data to main memory **606**, from which processor(s) **604** retrieves and executes the software instructions. The software instructions received by main memory **606** may optionally be stored on storage device(s) **610** either before or after execution by processor(s) **604**.

(125) Computing device **600** also may include one or more communication interface(s) **618** coupled to bus **602**. A communication interface **618** provides a two-way data communication coupling to a wired or wireless network link **620** that is connected to a local network **622** (e.g.,

Ethernet network, Wireless Local Area Network, cellular phone network, Bluetooth wireless network, or the like). Communication interface **618** sends and receives electrical, electromagnetic, or optical signals that carry digital data streams representing various types of information. For example, communication interface **618** may be a wired network interface card, a wireless network interface card with an integrated radio antenna, or a modem (e.g., ISDN, DSL, or cable modem). (126) Network link(s) **620** typically provide data communication through one or more networks to other data devices. For example, a network link **620** may provide a connection through a local network **622** to a host computer **624** or to data equipment operated by an Internet Service Provider (ISP) **626**. ISP **626** in turn provides data communication services through the world wide packet data communication network now commonly referred to as the “Internet” **628**. Local network(s) **622** and Internet **628** use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link(s) **620** and through communication interface(s) **618**, which carry the digital data to and from computing device **600**, are example forms of transmission media.

(127) Computing device **600** can send messages and receive data, including program code, through the network(s), network link(s) **620** and communication interface(s) **618**. In the Internet example, a server **630** might transmit a requested code for an application program through Internet **628**, ISP **626**, local network(s) **622** and communication interface(s) **618**.

(128) The received code may be executed by processor **604** as it is received, and/or stored in storage device **610**, or other non-volatile storage for later execution.

Basic Software System

(129) FIG. 7 is a block diagram of a basic software system **700** that may be employed for controlling the operation of computing device **600**. Software system **700** and its components, including their connections, relationships, and functions, is meant to be exemplary only, and not meant to limit implementations of the example embodiment(s). Other software systems suitable for implementing the example embodiment(s) may have different components, including components with different connections, relationships, and functions.

(130) Software system **700** is provided for directing the operation of computing device **600**. Software system **700**, which may be stored in system memory (RAM) **606** and on fixed storage (e.g., hard disk or flash memory) **610**, includes a kernel or operating system (OS) **710**.

(131) The OS **710** manages low-level aspects of computer operation, including managing execution of processes, memory allocation, file input and output (I/O), and device I/O. One or more application programs, represented as **702A**, **702B**, **702C** . . . **702N**, may be “loaded” (e.g., transferred from fixed storage **610** into memory **606**) for execution by the system **700**. The applications or other software intended for use on device **700** may also be stored as a set of downloadable computer-executable instructions, for example, for downloading and installation from an Internet location (e.g., a Web server, an app store, or other online service).

(132) Software system **700** includes a graphical user interface (GUI) **715**, for receiving user commands and data in a graphical (e.g., “point-and-click” or “touch gesture”) fashion. These inputs, in turn, may be acted upon by the system **700** in accordance with instructions from operating system **710** and/or application(s) **702**. The GUI **715** also serves to display the results of operation from the OS **710** and application(s) **702**, whereupon the user may supply additional inputs or terminate the session (e.g., log off).

(133) OS **710** can execute directly on the bare hardware **720** (e.g., processor(s) **604**) of device **600**. Alternatively, a hypervisor or virtual machine monitor (VMM) **730** may be interposed between the bare hardware **720** and the OS **710**. In this configuration, VMM **730** acts as a software “cushion” or virtualization layer between the OS **710** and the bare hardware **720** of the device **600**.

(134) VMM **730** instantiates and runs one or more virtual machine instances (“guest machines”). Each guest machine comprises a “guest” operating system, such as OS **710**, and one or more applications, such as application(s) **702**, designed to execute on the guest operating system. The

VMM 730 presents the guest operating systems with a virtual operating platform and manages the execution of the guest operating systems.

(135) In some instances, the VMM 730 may allow a guest operating system to run as if it is running on the bare hardware 720 of device 600 directly. In these instances, the same Revision of the guest operating system configured to execute on the bare hardware 720 directly may also execute on VMM 730 without modification or reconfiguration. In other words, VMM 730 may provide full hardware and CPU virtualization to a guest operating system in some instances.

(136) In other instances, a guest operating system may be specially designed or configured to execute on VMM 730 for efficiency. In these instances, the guest operating system is “aware” that it executes on a virtual machine monitor. In other words, VMM 730 may provide para-virtualization to a guest operating system in some instances.

(137) The above-described basic computer hardware and software is presented for purpose of illustrating the basic underlying computer components that may be employed for implementing the example embodiment(s). The example embodiment(s), however, are not necessarily limited to any particular computing environment or computing device configuration. Instead, the example embodiment(s) may be implemented in any type of system architecture or processing environment that one skilled in the art, in light of this disclosure, would understand as capable of supporting the features and functions of the example embodiment(s) presented herein.

Extensions and Alternatives

(138) In the foregoing specification, the example embodiment(s) of the present invention have been described with reference to numerous specific details. However, the details may vary from implementation to implementation according to the requirements of the particular implement at hand. The example embodiment(s) are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A method of data revision control for a distributed file system, comprising: managing via a build service versioning of a plurality of datasets stored in the distributed file system, wherein each version of a dataset of the plurality of datasets is transformed by interpreting database schema information in a committed database transaction; storing first transaction data for a first transaction producing a first version of a first dataset, second transaction data for a second transaction transforming the first version of the first dataset to a second version of the first dataset, and third transaction data for a third transaction transforming the second version of the first dataset to a third version of the first dataset; storing dependency information indicating a first dependency of the third transaction producing the third version of the first dataset on the second transaction producing the second version of the first dataset and a second dependency of the second transaction on the first transaction producing the first version of the first dataset; receiving a first instruction to revoke a first permission of a first user to access the second version of the first dataset; automatically revoking, in response to the receiving, a certain permission of the first user to access the third version of the first dataset based on the dependency information, wherein the method is performed by one or more processors.

2. The method of claim 1, further comprising: receiving a second instruction to revoke a second permission of a second user to access the first version of the first dataset; automatically revoking a particular permission of the second user to access the third version of the first dataset based on the first dependency and the second dependency.

3. The method of claim 1, further comprising: receiving a second instruction to grant a second permission of a second user to access the second version of the first dataset; automatically granting a particular permission of the second user to access the third version of the first dataset based on the dependency information.

4. The method of claim 1, further comprising: storing fourth transaction data for a fourth transaction producing a first version of a second dataset and fifth transaction data for a fifth transaction transforming a first column of the second version of the first dataset and a first column of the first version of the second dataset to a first column of a first version of a third dataset; storing further dependency information indicating a third dependency of the fifth transaction on the second transaction and the fourth transaction.
5. The method of claim 4, further comprising: receiving a second instruction to revoke a second permission of the first user to access the first column of the second version of the first dataset; automatically revoking a particular permission of the first user to access the first column of the first version of the third dataset.
6. The method of claim 4, further comprising: receiving a second instruction to grant a second permission of a second user to access the first column of the second version of the first dataset; automatically granting a particular permission of the second user to access the first column of the first version of the third dataset.
7. The method of claim 4, further comprising: receiving a second instruction to revoke a second permission of the first user to access a first column of the first version of the first dataset; automatically revoking a particular permission of the first user to access the first column of the first version of the third dataset.
8. The method of claim 1, further comprising: parsing instructions of a driver program, which when executed on a client causes applying parallel transformation operations at a plurality of workers to versions of the plurality of datasets; identifying, based on the parsing, a transformation operation, a first version of a second dataset as an input dataset of the transformation operation, and a first version of a third dataset as an output dataset of the transformation operation.
9. The method of claim 8, further comprising: obtaining schemas of the first version of the second dataset and the first version of the third dataset; determining, based on the obtained schemas, that a first column of the first version of the second dataset affects a first column of the first version of the third dataset; storing, in a database, column provenance information for the first column of the first version of the third dataset.
10. The method of claim 9, further comprising causing displaying, in a user interface, a column provenance graph having a directed link from the first column of the first version of the second dataset to the first column of the first version of the third dataset based on the column provenance information.
11. One or more non-transitory, computer-readable storage media storing one or more sequences of instructions which, when executed cause one or more processors to perform: managing via a build service versioning of a plurality of datasets stored in a distributed file system, wherein each version of a dataset of the plurality of datasets is transformed by interpreting database schema information in a committed database transaction; storing first transaction data for a first transaction producing a first version of a first dataset, second transaction data for a second transaction transforming a first version of the first dataset to a second version of the first dataset, and third transaction data for a third transaction transforming the second version of the first dataset to a third version of the first dataset; storing dependency information indicating a first dependency of the third transaction producing the third version of the first dataset on the second transaction producing the second version of the first dataset and a second dependency of the second transaction on the first transaction producing the first version of the first dataset; receiving a first instruction to revoke a first permission of a first user to access the second version of the first dataset; automatically revoking, in response to the receiving, a certain permission of the first user to access the third version of the first dataset based on the dependency information.
12. The one or more non-transitory, computer-readable storage media of claim 11, the one or more sequences of instructions further causing the one or more processors to perform: receiving a second instruction to revoke a second permission of a second user to access the first version of the first

dataset; automatically revoking a particular permission of the second user to access the third version of the first dataset based on the first dependency and the second dependency.

13. The one or more non-transitory, computer-readable storage media of claim 11, the one or more sequences of instructions further causing the one or more processors to perform: receiving a second instruction to grant a second permission of a second user to access the second version of the first dataset; automatically granting a particular permission of the second user to access the third version of the first dataset based on the dependency information.

14. The one or more non-transitory, computer-readable storage media of claim 11, the one or more sequences of instructions further causing the one or more processors to perform storing fourth transaction data for a fourth transaction producing a first version of a second dataset and fifth transaction data for a fifth transaction transforming a first column of the second version of the first dataset and a first column of the first version of the second dataset to a first column of a first version of a third dataset; storing further dependency information indicating a third dependency of the fifth transaction on the second transaction and the fourth transaction.

15. The one or more non-transitory, computer-readable storage media of claim 14, the one or more sequences of instructions further causing the one or more processors to perform: receiving a second instruction to revoke a second permission of the first user to access the first column of the second version of the first dataset; automatically revoking a particular permission of the first user to access the first column of the first version of the third dataset.

16. The one or more non-transitory, computer-readable storage media of claim 14, the one or more sequences of instructions further causing the one or more processors to perform, receiving a second instruction to grant a second permission of a second user to access the first column of the second version of the first dataset; automatically granting a particular permission of the second user to access the first column of the first version of the third dataset.

17. The one or more non-transitory, computer-readable storage media of claim 14, the one or more sequences of instructions further causing the one or more processors to perform: receiving a second instruction to revoke a second permission of the first user to access a first column of the first version of the first dataset; automatically revoking a particular permission of the first user to access the first column of the first version of the third dataset.

18. The one or more non-transitory, computer-readable storage media of claim 11, the one or more sequences of instructions further causing the one or more processors to perform parsing instructions of a driver program, which when executed on a client causes applying parallel transformation operations at a plurality of workers to versions of the plurality of datasets; identifying, based on the parsing, a transformation operation, a first version of a second dataset as an input dataset of the transformation operation, and a first version of a third dataset as an output dataset of the transformation operation.

19. The one or more non-transitory, computer-readable storage media of claim 18, the one or more sequences of instructions further causing the one or more processors to perform: obtaining schemas of the first version of the second dataset and the first version of the third dataset; determining, based on the obtained schemas, that the first column of the first version of the second dataset affects the first column of the first version of the third dataset; storing, in a database, column provenance information for the first column of the first version of the third dataset.

20. The one or more non-transitory, computer-readable storage media of claim 19, the one or more sequences of instructions further causing the one or more processors to perform causing displaying, in a user interface, a column provenance graph having a directed link from the first column of the first version of the second dataset to the first column of the first version of the third dataset based on the column provenance information.
